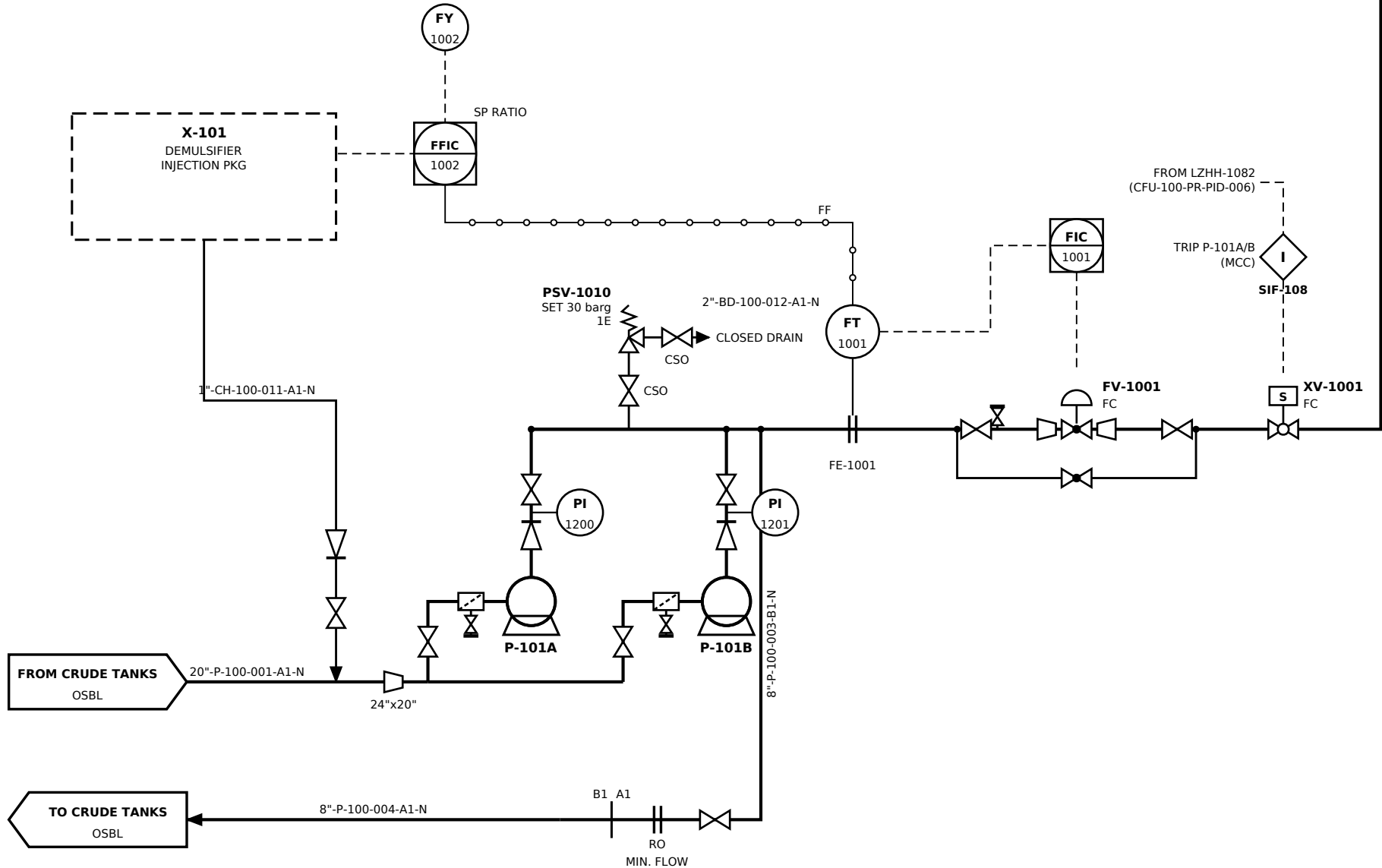
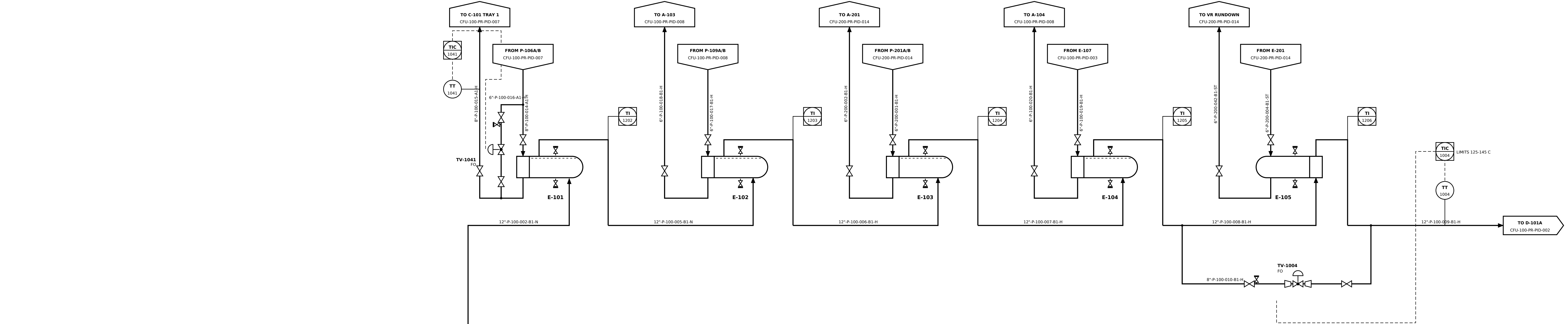


|                                                                                                                                              |                                                                                                                                                                                                    |                                                                                                                                                                                                        |                                                                                                                                                                                                                       |                                                                                                                                                                                                    |                                                                                                                                                                                                           |                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>P-101A/B</b><br>Crude charge<br>RATED 730 m3/h @ 261 m, @P 22 bar<br>MOTOR 710 kW, API 682, Flm 11/22<br>DES. T 60 C; MAT'L S-6 (CS/12Cr) | <b>E-101</b><br>Crude / Top pumparound<br>359 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 185 C; CA 3 mm<br>DUTY 8.09 MW<br>SHELL: Crude / TUBE: Top pumparound<br>MAT'L CS shell / CS tubes | <b>E-102</b><br>Crude / Kerosene product<br>438 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 240 C; CA 3 mm<br>DUTY 6.55 MW<br>SHELL: Crude / TUBE: Kerosene product<br>MAT'L CS shell / CS tubes | <b>E-103</b><br>Crude / LVGO pumparound/product<br>804 m2 total; 25 x 1P shells<br>DESIGN 35 / 20 barg / 255 C; CA 3 mm<br>DUTY 11.45 MW<br>SHELL: Crude / TUBE: LVGO pumparound/product<br>MAT'L CS shell / CS tubes | <b>E-104</b><br>Crude / Diesel product<br>378 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 235 C; CA 3 mm<br>DUTY 4.44 MW<br>SHELL: Crude / TUBE: Diesel product<br>MAT'L CS shell / CS tubes | <b>E-105</b><br>Crude / Vacuum residue<br>431 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 285 C; CA 3 mm<br>DUTY 5.84 MW<br>SHELL: Vacuum residue / TUBE: Crude<br>MAT'L CS shell / 5Cr-1/2Mo tubes | <b>X-101</b><br>Demulsifier injection package<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR 0.75 kW<br>MAT'L SS316 |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|



- NOTES**
1. Cold-train crude-side design 35 barg (E-101, E-105) >= P-101 shut-off; PSV-1010 (30 barg) thermal / blocked-in.
  2. Exchanger hot-side connections shown at channel / shell nozzles; TEMA type & shells per data sheets.
  3. TV-1041 / TV-1004 bypass control valves have block valves only (exchanger provides bypass path).
  4. Demulsifier dosed at P-101 suction, ratio to FIC-1001 (FFIC-1002) - X-101 vendor package.
  5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN    |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 2 OF 17    |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-100-PR-PID-001                         |            |                          | A          |